Paper - 70% | Assignments - 30%

No. ..... Date. 22/06/2026.

Artificial Intelligence

Content

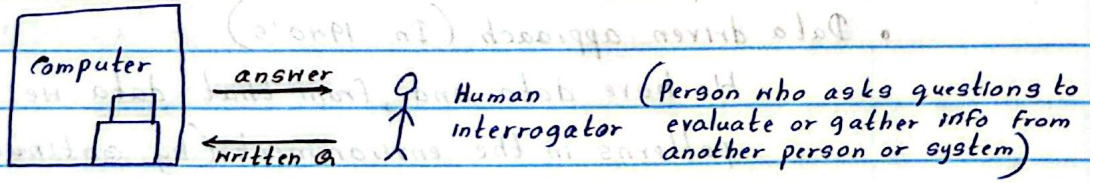
01- History of AI
02- Agents
03- Search - To solve problems. When AI is applied to a problem there is always a search. There is optimization in any AI application.
04- Genetic Algorithms (GA) - Extended version of search.
05- Machine Learning (ML)
06- Artificial Neural Networks - dominant AI approach today } can be hybridized LCS (Learning Classifier Systems)
07- Rule based AI systems - currently known as GOFAI (Good Old Fashion AI) }
08- Fuzzy Logic Systems (Binary Linguistic Rules)
09- Swarm Intelligence
10- Basics of Reinforcement Learning

* AI is the science of developing techniques to generate intelligence artificially.

Inventors of AI → Marvin Minsky
John McCarthy

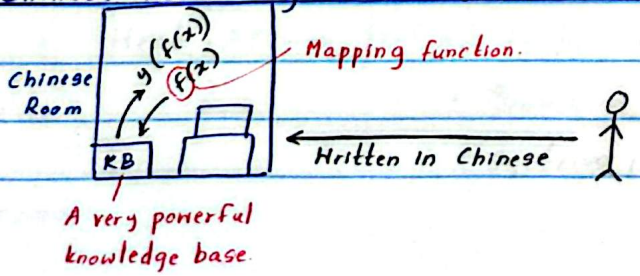
1951 - Successful architecture of Neuron.

~~Turnt~~ Turin test



* The human cannot see inside. He gives written questions. If those humans cannot identify it is a computer, the machine is intelligent.

Chinese Room Argument - John Sherely.



* Get the question, translated and get the answer. Mapping process. Even the computer is not intelligent It can act like intelligent.

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AI discipline → 1956 - Making a machine that can think like a human.

* Two scientists Marrin & John called a conference and this was a 9 months discussion of making a mechanical brain. This conference is called 'Dartmouth Conference' and the main purpose was to talk how AI is to be build. The term AI was also introduced by them.

In this conference they decide that symbol processing is the best way of creating the brain.

eg: if.... then.... (symbol processing)

- Rule based approach - Facts and rules are made by symbols.

eg: If temperature > 38 C, then patient has a ferer.

(GOFAI - Good Old Fashion AI)

Rule based programming languages

- Prolog ~~(To imp~~
- LISP - not rule based by default, but can be used for rule based AI

(To implement this McCarthy created the language LISP)

- Data driven approach (In 1940's)

We have data and, from that data we identify certain patterns in the environment. (by continuous feed of data)

* Today rule based approach is called GOFAI and data driven approach is called Machine Learning.

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Disruptive technologies

- A disruptive technology is a technology that changes the normal way things are done & replaces older methods.

Ex. of disruptive technologies.

- AI
- Smartphones
- Cloud computing
- Block chain
- Internet

* Centralization is the base of modern technology. This concept was dropped and blockchain came to use which is a desruptive technology with the concept of decentralization.

* However blockchain introduced decentralization as an alternative. Both systems coexist based on use case.

Why we need to study AI ?

To face the disruptive technologies and create a new product and tool. AI can give better solutions for problems we face. (optimization)

* Humans - Carbon brain

AI - Silicone brain

Strong AI ⟶ Machine is truely thinking as a human brain. Functionality of a real human brain.
- Chatbots (NLP), Autonomous vehicles.

Weak AI ⟶ Pretend to be intelligent. Having a function.
- Browsing the internet, Disease detection., Recommender systems in Amazon.

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Main areas of AI

- ANI. (Artificial Narrow Intelligence)
- AGI (Artificial General Intelligence)
- Artificial Super Intelligence (ASI)

01- Artificial Narrow Intelligence.

- AI that designed to perform one specific task or a limited set of tasks.
  - * Siri / Alexa (voice assistants)
  - * Google Maps.
  - * Face recognition systems.

02- Artificial General Intelligence.

- AI that can think, learn and perform any intellectual task like a human. Can learn new tasks without retraining.

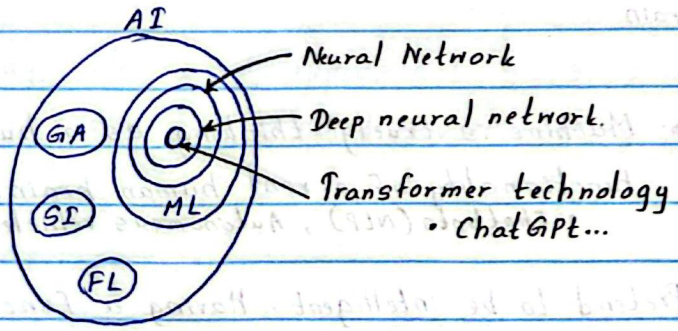
Not yet achieved - hypothetical ex.

- A robot that can study, cook, drive & solve problems like a human.

03- Artificial Super Intelligence.

- AI that is far more intelligent than humans in every field.

Future concept - AI that can outperform humans in science, creativity, decision making & emotions.



AI from the lense of science.

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AI

Discriminative AI

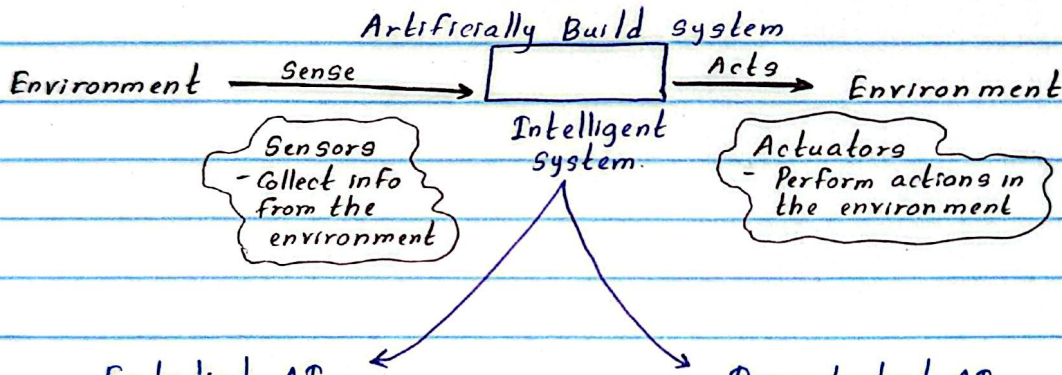
- Used to classify or predict something based on input data. (Predictive AI)

ex: * Email spam detection.
* Face recognition.

Generative AI

- Used to create new content like images, text, audio, video. (Gen AI)

ex: * ChatGPt
* Music generation AI.



Embodied AI

- AI residing in physical body
  - Self driving cars
  - Humanoid robots

Disembodied AI

- AI * residing in only a software component.
  - ChatGPt
  - Google translate.

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